

Parallel Portfolio Backtests by Global Newton Iteration

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Abstract

Portfolio backtests with trading costs and turnover constraints couple decisions across dates. This requires updating the portfolio sequentially across dates to obtain the holdings trajectory and evaluate portfolio performance. Given the GPU resources available today, a procedure that is parallelizable across dates and converges to the same holdings trajectory offers the potential for significant speedup. We evaluate a fixed portfolio policy in parallel using global Newton iteration, parallel daily optimizations, and affine scans. Nine policies are studied on 491 stocks over 252 trading days. The state consists of dollar holdings and cash, with separate step sizes at each date. With an allowance of at most 50 batches, maximum NAV errors are 0.0001412% for the plain long-only policy and 6.552% for the plain long-short policy; annualized return tracking errors are $1.148 \times 10^{-5}\%$ and 2.858%, respectively. Sequential and parallel portfolio metrics are reported alongside recurrence residuals and accounting gaps.

1 Introduction

A portfolio backtest repeatedly constructs a portfolio, executes its trades, pays transaction costs, and applies the next period’s market returns. With turnover limits and trading costs, each construction depends on the holdings entering that date. Even when the market data and forecasts are fixed, the usual implementation therefore solves one portfolio problem after another, i.e., sequentially across dates.

We study a parallel alternative that treats the entire sequence of holdings and cash balances as unknowns. Daily portfolio problems are evaluated concurrently at a candidate sequence. Their sensitivities are then combined to update that sequence using Newton’s method. The parallel procedure aims to replicate the sequence of holdings obtained by running the sequential backtest. The sequential result is used only to measure accuracy; it is not used to initialize, update, or stop the parallel run.

The practical question is how closely a limited number of parallel iterations reproduces a sequential backtest. We examine dollar holdings, NAV and daily returns, together with the residual of the underlying accounting recurrence.

This is an empirical study of an implemented procedure.

2 Related work

Convex portfolio construction combines expected return, risk, transaction costs, and trading constraints. Boyd et al. [1] describe a unified single-period and multi-period trading framework. Our daily policy is a single-period convex optimization followed by explicit dollar accounting. We study the numerical evaluation of this fixed policy through historical time.

Parallel prefix computations are classical; Blelloch [2] provides the underlying scan framework. Lim et al. [3] apply Newton linearization and parallel recurrence solves to general nonlinear sequential models in their DEER (nonlinear differential equation as fixed-point iteration) framework. The global

Newton/affine-scan mechanism is therefore established. Here the transition contains a constrained optimizer, nonsmooth trading costs, and a wealth-dependent impact model, which require additional sensitivity and feasibility analysis.

Implicit differentiation of convex optimization solution maps is developed, for example, by Agrawal et al. [4]. We differentiate a selected active Karush–Kuhn–Tucker (KKT) system, including the curvature of active risk and cash constraints. Clarabel [5] solves the daily conic problems. At changes in the active constraints or trade signs, the selected sensitivity can be less accurate.

3 Portfolio policy and accounting

Let $s_t = (h_t, c_t) \in \mathbb{R}^{n+1}$, $t = 1, \dots, T$, denote risky dollar holdings and cash entering date t , with s_0 the initial state. Let $V = \mathbf{1}^\top s_t > 0$ be the incoming wealth. Set $p = h_t/V$. The decision x is post-trade risky holdings divided by *incoming* wealth; $z = x - p$ is the trade in the same units. With daily covariance Σ_t , expected risky return μ_t , half spreads a_t , and dollar average daily volume ADV_{ti} , define the transaction cost as a fraction of incoming wealth, $C_t(z, V)$, by

$$C_t(z, V) = a_t^\top |z| + \eta V \sum_i z_i^2 / \text{ADV}_{ti}. \quad (1)$$

The first cost term represents the bid–ask spread and the second market impact; $VC_t(z, V)$ is the dollar cost. Cash earns zero return throughout.

The objective of each convex optimization policy studied is

$$\max_x \quad \mu_t^\top x - \frac{\gamma}{2} x^\top \Sigma_t x - \lambda C_t(z, V). \quad (2)$$

All policies impose nonnegative executed cash and separate buy and sell limits:

$$1 - \mathbf{1}^\top x - C_t \geq 0, \quad \mathbf{1}^\top (z)_+ \leq \tau, \quad \mathbf{1}^\top (-z)_+ \leq \tau. \quad (3)$$

Thus turnover is the larger of total purchases and sales, divided by incoming wealth. Long-only policies impose $0 \leq x_i \leq 0.05$ and $\|x\|_1 \leq 1$. Long-short policies impose $|x_i| \leq 0.05$, $\|x\|_1 \leq 1.5$, and $\mathbf{1}^\top x = 0$. Thus their long and short dollar positions balance immediately after construction; subsequent market returns can change the net exposure. Optional constraints are $\sqrt{252 x^\top \Sigma_t x} \leq 0.10$ and $\ell \leq B_t^\top x \leq b$, where the columns of B_t contain asset exposures. For beta, the benchmark holds equal weights $w = \mathbf{1}/n$ in the eligible stocks. Asset i 's beta is its covariance with that benchmark return divided by the benchmark's variance:

$$\beta_{ti} = \frac{(\Sigma_t w)_i}{w^\top \Sigma_t w}, \quad \beta_t^{\text{portfolio}} = \beta_t^\top x. \quad (4)$$

Thus beta measures sensitivity to movements in the equal-weight benchmark return. Sector columns contain one for assets in that sector and zero otherwise. The covariance uses the factor representation $\Sigma_t = L_t L_t^\top + \text{diag}(d_t^2)$, where L_t is the matrix of scaled factor loadings and d_t contains idiosyncratic standard deviations.

For realized risky returns r_t , write $R_t = \text{diag}(1 + r_t, 1)$. The complete transition is

$$F_t(s_t) = R_t \begin{bmatrix} Vx \\ V - \mathbf{1}^\top Vx - VC_t \end{bmatrix}, \quad s_{t+1} = F_t(s_t), \quad s_0 \text{ fixed}. \quad (5)$$

Dollar costs equal $a_t^\top |Vx - h_t| + \eta \sum_i (Vx_i - h_{ti})^2 / \text{ADV}_{ti}$. Costs are paid before returns accrue, so executed wealth is exactly $V - VC_t$. Short holdings remain signed; cash cannot be borrowed. The model does not charge stock-borrow fees. Forecasts and eligible assets are fixed before the recurrence solve; there is no forecast adaptation to Newton iterates. Wealth-dependent impact couples weights and dollars.

4 Global Newton with separate date step sizes

Set $u_t = s_0$ for $0 \leq t \leq T$. Let Q count paid evaluation batches and $Q_{\max}$ denote the batch allowance. A paid batch evaluates all $F_t(u_t)$ and selected sensitivities A_t at the current path. On a regular differentiable active-constraint face, these are the Jacobians of the complete transition:

$$A_t = \left. \frac{\partial F_t(s)}{\partial s} \right|_{s=u_t} \in \mathbb{R}^{(n+1) \times (n+1)}, \quad (A_t)_{ij} = \left. \frac{\partial F_{t,i}}{\partial s_j} \right|_{s=u_t}. \quad (6)$$

At a kink, the implementation differentiates a selected active KKT system. For long-short policies this system includes the equality $\mathbf{1}^\top x = 0$ and its tangent $\mathbf{1}^\top dx = 0$. Define

$$q_t = F_t(u_t) - u_{t+1}, \quad \epsilon_R(u) = \max_{0 \leq t < T} \|q_t\|_1 / V_0, \quad V_0 = \mathbf{1}^\top s_0. \quad (7)$$

The algorithm is as follows. Return with converged status only if $\epsilon_R \leq 10^{-5}$. Otherwise solve

$$\delta u_0 = 0, \quad \delta u_{t+1} = A_t \delta u_t + q_t. \quad (8)$$

The affine scan solves this recurrence as follows. For adjoining intervals L then R , affine composition is

$$(A_R, q_R) \circ (A_L, q_L) = (A_R A_L, A_R q_L + q_R). \quad (9)$$

An upward tree pass forms interval maps, and a downward pass recovers every correction. The composition order is essential because matrices need not commute. The implementation uses $T - 1$ dense matrix products, $O(Tn^2)$ additional vector work, and two passes of $\lceil \log_2 T \rceil$ levels. These are arithmetic depth statements, not measured speedups.

Dollar-state update. At each date, use a separate step size

$$\alpha_t = \min \left(1, \frac{0.25 \mathbf{1}^\top u_t}{\max(\|\delta u_t\|_1, 10^{-30})} \right), \quad v_t = u_t + \alpha_t \delta u_t. \quad (10)$$

Keep $v_0 = s_0$. Clip negative cash and, for long-only policies, negative risky holdings; reject nonfinite states or nonpositive wealth. These are domain checks, not a projection onto post-trade position or turnover constraints. Realized returns can legitimately move a portfolio beyond construction limits. Clipping can change NAV, so we report the resulting accounting gap.

Evaluate the date-capped proposal v_t at every date and accept it if all policy evaluations and sensitivities succeed. The checks require an optimal solver status, finite states with positive NAV, feasibility of executed holdings, cash, turnover and any risk or exposure constraints to the post-solve numerical tolerance, and finite sensitivity matrices. For long-short policies, the feasibility check also enforces dollar neutrality. If accepted, set $u_t = v_t$ at every date and increase the batch count by one; the allowance is $Q_{\max}$ batches. Portfolio evaluations within a batch can run in parallel, while successive batches must be processed sequentially. After a failed proposal, evaluate one simultaneous fallback $v_0 = s_0$, $v_{t+1} = F_t(u_t)$ using the retained outputs and increase the batch count by 1. This is a parallel fixed-point step. Stop if it also fails or the paid budget is exhausted. Clarabel's absolute-gap, relative-gap, and feasibility tolerances are 10^{-7} . The post-solve check allows violations up to 2×10^{-5} in normalized holdings, cash, turnover, and exposures, and in annualized variance.

Initialization, rejected trials, and fallbacks all consume paid batches Q . Every full batch makes T policy calls. Only the last accepted, evaluated path is returned. After Q paid batches with no failures, there are $Q - 1$ accepted updates and scans; failed trials can make the accepted update count smaller. The convergence flag is determined by the original dollar residual ϵ_R .

5 Experiments and quantitative diagnostics

5.1 Data and protocol

The study contains 252 transitions, 9 January 2023–10 January 2024, and 491 eligible stocks. Initial wealth is \$1 million in cash. Expected returns use cross-sectionally standardized trailing 12–1 log-momentum scores, clipped to $[-3, 3]$ and scaled by $0.10/252$. Covariances use 252-day trailing windows of volatility-standardized, clipped returns, 20 factors, and at most 10 expectation–maximization (EM) iterations. EM alternates between estimating latent factor moments and updating the factor loadings and idiosyncratic variances; see Tzikas et al. [6] for related statistical risk modeling. ADV uses 20 days; half spreads are 5 basis points; $\eta = 0.1$, $\gamma = 5$, and $\lambda = 1$.

All nine one-year policies use $\tau = 10\%$. The hard-risk-only variant sets $\gamma = 0$; the combined risk variants retain $\gamma = 5$. Beta ranges are $[0, 0.5]$ for long-only and $[-0.15, 0.15]$ for signed policies. Three exposures means beta plus technology and financial sector caps of 20% each. Sector caps means beta plus all 11 sector caps of 20%. The membership snapshots are current constituent lists, not historical point-in-time membership. Consequently survivorship and selection bias affect these historical returns. The experiments assess numerical reconciliation; they are not evidence of an investable forecasting advantage.

In the main experiment, each parallel run starts from the constant all-cash path and uses a paid allowance of $Q_{\max} = 50$. Runs stop earlier if the recurrence residual meets the stopping tolerance. Sequential references enter diagnostic callbacks only. Comparisons use the last accepted state within that allowance; rejected trials are charged. The implementation uses CPU worker processes; policy depth measures dependence across dates, not GPU runtime.

5.2 Finite-budget convergence and portfolio agreement

Table 1 reports the retained method at a common paid allowance. For candidate NAV $N_t = \mathbf{1}^\top u_t$, the daily local net return is

$$r_t^{\text{loc}} = \frac{\mathbf{1}^\top F_t(u_t)}{N_t} - 1.$$

We compute $F_t(u_t)$ by solving the portfolio policy at u_t , deducting transaction costs, and applying the realized asset returns. Thus r_t^{loc} can differ from $N_{t+1}/N_t - 1$ when $u_{t+1} \neq F_t(u_t)$. TE is $\sqrt{252}$ times the sample standard deviation of the differences between independently evaluated daily net returns. TE centers the return differences and therefore does not measure their mean bias. NAV error is $\max_t |N_t/N_t^{\text{seq}} - 1|$, where N_t^{seq} is sequential NAV. For compounded local-return wealth $W_t = V_0 \prod_{j < t} (1 + r_j^{\text{loc}})$, the accounting gap is $\max_t |N_t - W_t|/V_0$. This gap measures the largest difference between the wealth recorded in the candidate states and the wealth obtained by compounding the returns of the portfolios evaluated at those states, relative to initial wealth.

Policy	NAV error (%)	TE (%)	Gap (%)	ϵ_R
Long only	0.0001412	1.148×10^{-5}	0.0001805	2.071×10^{-6}
Hard risk	0.0001977	8.01×10^{-5}	0.0001202	1.064×10^{-6}
Beta cap	0.0001972	9.113×10^{-5}	0.0002307	4.276×10^{-6}
Risk + beta	0.000216	2.448×10^{-5}	0.0002433	2.042×10^{-6}
Three exposures	0.0001827	3.386×10^{-5}	0.0002534	3.444×10^{-6}
Sector caps	0.0001935	6.046×10^{-5}	0.0001721	9.272×10^{-6}
Long short	6.552	2.858	0.8892	0.2296
Signed + beta	3.353	3.462	1.373	0.225
Signed + risk + beta	2.961	2.753	2.516	0.2806

Table 1: Parallel Newton accuracy at a paid allowance of 50. NAV error, TE and gap are percentages; the recurrence residual is a fraction of initial NAV. All policies use objective (2) with $\lambda = 1$ and $\eta = 0.1$, nonnegative executed cash, and separate buy and sell limits $\tau = 0.10$. **Long only**: $0 \leq x_i \leq 0.05$, $\|x\|_1 \leq 1$, and $\gamma = 5$. **Hard risk**: long only with $\gamma = 0$ and annualized volatility at most 10%. **Beta cap**: long only with $0 \leq \beta_t^\top x \leq 0.5$. **Risk + beta**: beta cap with annualized volatility at most 10% and $\gamma = 5$. **Three exposures**: beta cap with Information Technology and Financials sector weights each in $[0, 0.2]$. **Sector caps**: beta cap with every one of the 11 GICS sector weights at most 0.2 (nonnegativity follows from long-only holdings). **Long short**: $|x_i| \leq 0.05$, $\|x\|_1 \leq 1.5$, $\mathbf{1}^\top x = 0$, and $\gamma = 5$. **Signed + beta**: long short with $-0.15 \leq \beta_t^\top x \leq 0.15$. **Signed + risk + beta**: signed + beta with annualized volatility at most 10%. Only the stated variants impose volatility or exposure bounds; “signed” means long-short.

Across the nine policies, maximum NAV error ranges from 0.0001412% to 6.552%. The plain long-short policy has NAV error 6.552%, compared with 0.0001412% for long-only.

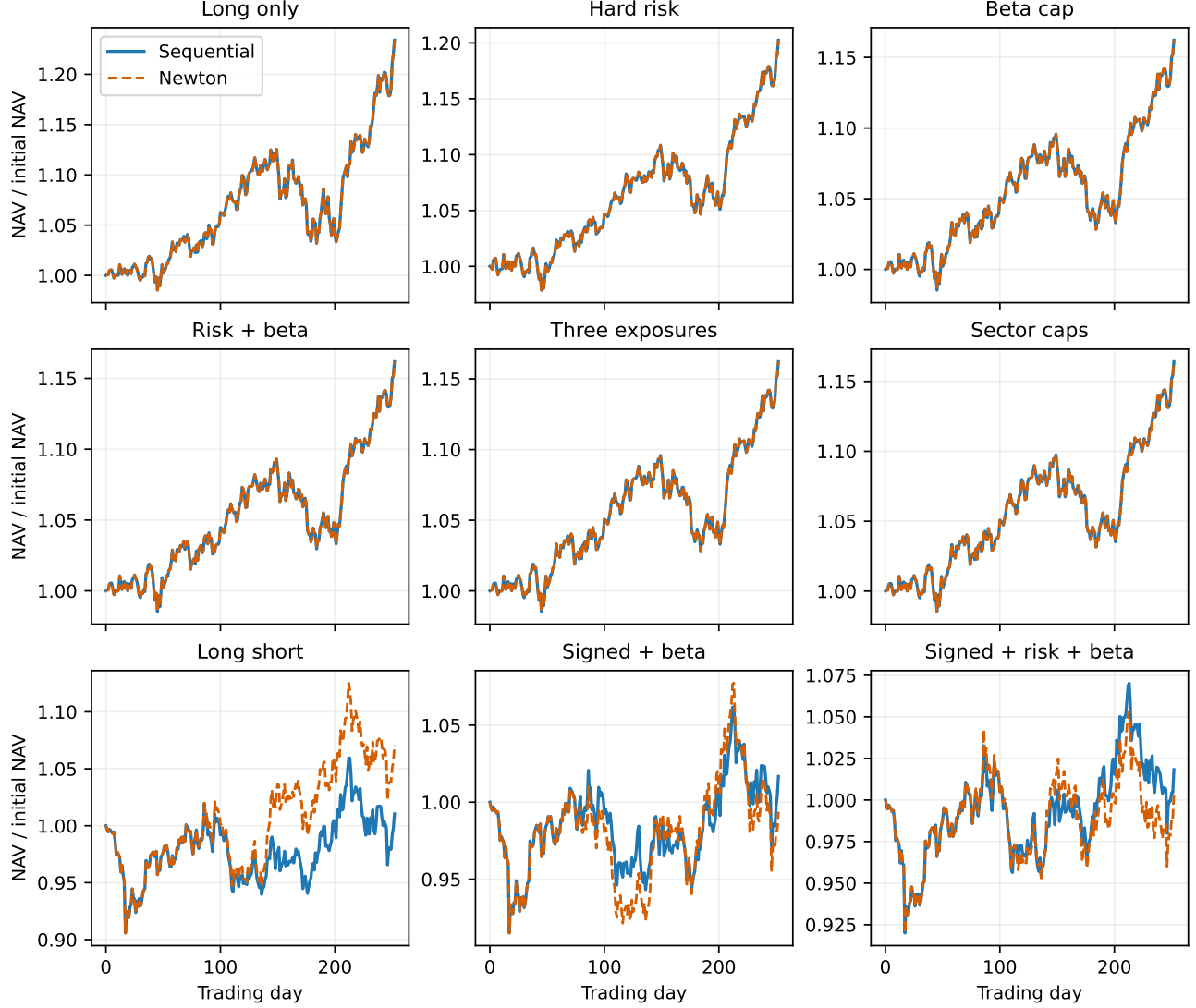


Figure 1: NAV trajectories for sequential backtesting and parallel Newton at the final allowance of 50 batches, for all nine policies. Each panel shows NAV relative to the common initial wealth.

The maximum nonlinear policy depth is $50/252 = 19.84\%$ of sequential depth. Total policy work is nevertheless up to 50 times that of a sequential backtest, plus sensitivity and dense scan work, but the policy evaluations are parallelizable within each batch. Figure 1 compares the final Newton and sequential NAV trajectories.

5.3 Sequential and parallel portfolio metrics

Table 2 retains the main portfolio outputs, placing each sequential reference beside its parallel candidate at the final allowance. CAGR, annualized volatility, and Sharpe use the evaluated local net returns, with sample standard deviations and zero cash return:

$$\text{CAGR} = \left[\prod_t (1 + r_t^{\text{loc}}) \right]^{252/T} - 1, \quad \text{Sharpe} = \sqrt{252} \frac{\text{mean}(r_t^{\text{loc}})}{\text{sd}(r_t^{\text{loc}})}.$$

NAV return and drawdown instead use the candidate states. This distinction matters when their accounting gap is nonzero. The sequential and parallel Sharpe ratios are useful comparisons, but agreement in Sharpe does not imply agreement in the entire portfolio path.

Policy	Method	NAV ret.	CAGR	Vol.	Sharpe	NAV DD
Long only	Sequential	23.40	23.40	12.07	1.803	-8.30
	Parallel	23.40	23.40	12.07	1.803	-8.30
Hard risk	Sequential	20.25	20.25	9.28	2.034	-5.57
	Parallel	20.25	20.25	9.28	2.034	-5.57
Beta cap	Sequential	16.22	16.22	8.60	1.792	-6.16
	Parallel	16.22	16.22	8.60	1.792	-6.16
Risk + beta	Sequential	16.18	16.18	8.37	1.835	-5.81
	Parallel	16.18	16.18	8.37	1.835	-5.81
Three exposures	Sequential	16.22	16.22	8.60	1.792	-6.16
	Parallel	16.22	16.22	8.60	1.792	-6.16
Sector caps	Sequential	16.41	16.41	8.58	1.814	-6.02
	Parallel	16.41	16.41	8.58	1.814	-6.02
Long short	Sequential	1.04	1.04	14.84	0.144	-9.43
	Parallel	7.08	6.85	15.21	0.512	-9.43
Signed + beta	Sequential	1.67	1.67	14.72	0.186	-8.52
	Parallel	-0.62	-0.26	15.20	0.059	-11.29
Signed + risk + beta	Sequential	1.83	1.83	13.20	0.203	-8.44
	Parallel	0.37	1.97	13.53	0.211	-8.83

Table 2: One-year portfolio metrics: sequential and parallel Newton at a paid allowance of 50. Returns, volatility, and drawdown are percentages. NAV metrics use states; CAGR, volatility, and Sharpe use local net returns.

For the plain long-short policy, sequential and parallel Sharpe ratios are 0.144 and 0.512. These are numerical discrepancies in evaluation of the same policy, not returns from different investment strategies.

6 Discussion and conclusion

Global Newton iteration evaluates the daily portfolio problems in parallel and uses one affine scan to update the full path. The experiments retain the same portfolio policy as the sequential reference and reduce the number of dependent policy evaluations, at the cost of more total work and dense matrix scans. At the studied budget, accuracy varies materially across portfolio constraints. Appendix A studies sequential three-day windows within each parallel batch. At a larger depth allowance, this lowers long-short return tracking errors, while its effects on NAV error depend on the constraints.

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A Sequential windows within Newton batches

We group successive dates into windows and apply the same Newton method to the window boundary states. Daily portfolio construction, transaction costs, and returns are evaluated sequentially within each window; different windows are evaluated in parallel. The policy, dated inputs, initial wealth, and sequential reference are the same as in the main experiments.

Let $0 = b_0 < b_1 < \dots < b_m = T$ be the boundaries, with at most W transitions per window, and let y_j be the candidate state at b_j . The window map is

$$G_j(y_j) = (F_{b_{j+1}-1} \circ \dots \circ F_{b_j})(y_j). \quad (\text{A.1})$$

Starting at y_j , we evaluate the daily transitions in chronological order. The selected window sensitivity is the chain-rule product

$$D_j = A_{b_{j+1}-1}^{(j)} \cdots A_{b_j}^{(j)}, \quad \delta y_0 = 0, \quad \delta y_{j+1} = D_j \delta y_j + G_j(y_j) - y_{j+1}, \quad (\text{A.2})$$

where each daily sensitivity is evaluated at the state propagated within that window. An affine scan solves the correction recurrence. We retain the separate 25% dollar-state caps, domain clipping, acceptance checks, and simultaneous fallback, now applied to boundary states. All boundary candidates start at the initial cash state, and $y_0 = s_0$ remains fixed.

For reporting, each window contributes its candidate starting state and cached sequential interior states; the final state is y_m . Thus the original daily recurrence residual is zero at interior joins and equals $G_j(y_j) - y_{j+1}$ at window boundaries. We apply the same 10^{-5} stopping tolerance to its maximum dollar ℓ_1 norm divided by initial wealth. Daily returns are those evaluated on this reconstructed path. Reconstruction requires no additional policy calls or sequential repair across windows.

We use $W = 3$, giving $m = 84$ windows, and a paid-batch allowance $Q_{\max} = 42$. The maximum dependent policy depth is

$$QW \leq 42 \cdot 3 = 126 = \frac{1}{2}T.$$

Initialization, rejected proposals, and fallback evaluations all count toward this allowance. A successful batch makes $T = 252$ daily policy calls. A failed window skips its dependent tail, but the full window depth is still charged. Daily sensitivities and their matrix products add work beyond these policy calls. The main paper uses at most 50 one-day batches, so its final results have a smaller depth allowance. Fewer window batches alone do not imply lower sequential depth or a measured speedup.

Window Newton reaches the stopping tolerance for 6 of the nine policies, compared with 6 for daily Newton. The long-only cases use 18–27 batches (policy depths 54–81). The long-short cases exhaust the window batch allowance. For the plain, beta-constrained, and risk-plus-beta long-short policies, respectively, final NAV errors are 1.318%, 0.6996%, 5.05%, and return tracking errors are 1.017%, 0.6085%, 1.707%. The corresponding daily-Newton tracking errors in the main paper are 2.858%, 3.462%, 2.753%. For risk plus beta, NAV error increases from 2.961% to 5.05%. These finite-budget return-tracking improvements do not certify convergence; the long-short recurrence residuals decrease in all three cases but remain above tolerance.

Policy	Q	QW	NAV error (%)	TE (%)	Gap (%)	ϵ_R
Long only	24	72	0.0006448	2.237×10^{-5}	0.0008244	9.917×10^{-6}
Hard risk	27	81	0.0002188	3.744×10^{-5}	0.0002506	6.751×10^{-6}
Beta cap	18	54	0.0002819	8.588×10^{-5}	0.0003035	8.272×10^{-6}
Risk + beta	19	57	7.539×10^{-5}	1.682×10^{-5}	8.756×10^{-5}	4.892×10^{-6}
Three exposures	24	72	0.0001705	2.39×10^{-5}	0.0002082	3.834×10^{-6}
Sector caps	24	72	0.0002494	6.612×10^{-5}	0.0001781	3.5×10^{-6}
Long short	42	126	1.318	1.017	0.5001	0.1419
Signed + beta	42	126	0.6996	0.6085	0.3684	0.06567
Signed + risk + beta	42	126	5.05	1.707	1.23	0.2127

Table A.1: Window Newton accuracy, corresponding to Table 1, with $W = 3$ and an allowance of 42 batches. Q counts paid batches and QW is charged policy depth. NAV error, TE, and accounting gap are percentages; ϵ_R is a fraction of initial NAV.

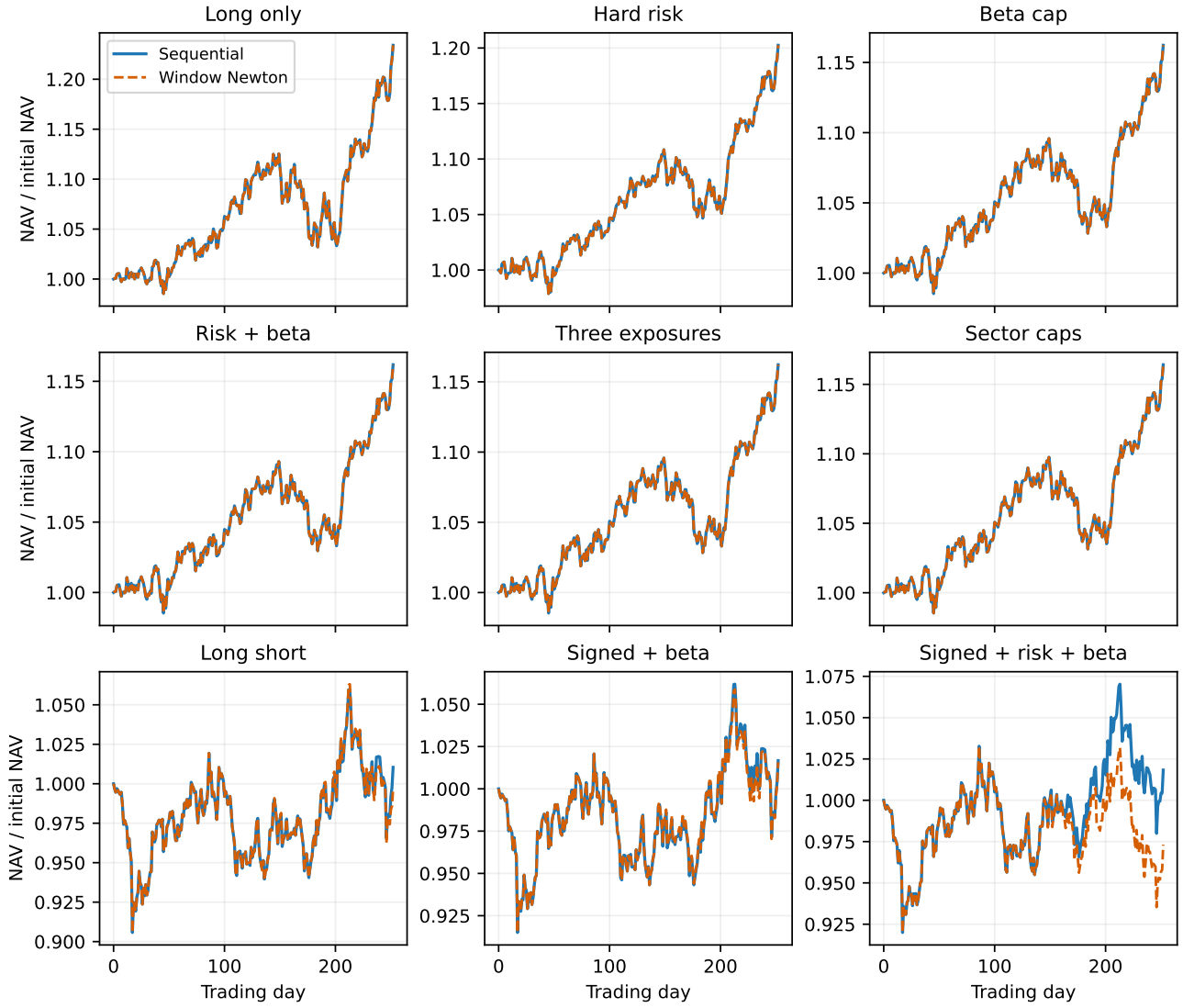


Figure A.1: Sequential and window Newton NAV trajectories, corresponding to Figure 1. Each window result is the last accepted path within $Q \leq 42$ and $QW \leq 126$, normalized by the common initial wealth.

Policy	Method	NAV ret.	CAGR	Vol.	Sharpe	NAV DD
Long only	Sequential	23.40	23.40	12.07	1.803	-8.30
	Window Newton	23.40	23.40	12.07	1.803	-8.30
Hard risk	Sequential	20.25	20.25	9.28	2.034	-5.57
	Window Newton	20.25	20.25	9.28	2.034	-5.57
Beta cap	Sequential	16.22	16.22	8.60	1.792	-6.16
	Window Newton	16.22	16.22	8.60	1.792	-6.16
Risk + beta	Sequential	16.18	16.18	8.37	1.835	-5.81
	Window Newton	16.18	16.18	8.37	1.835	-5.81
Three exposures	Sequential	16.22	16.22	8.60	1.792	-6.16
	Window Newton	16.22	16.22	8.60	1.792	-6.16
Sector caps	Sequential	16.41	16.41	8.58	1.814	-6.02
	Window Newton	16.41	16.41	8.58	1.814	-6.02
Long short	Sequential	1.04	1.04	14.84	0.144	-9.43
	Window Newton	-0.29	0.18	14.91	0.087	-9.47
Signed + beta	Sequential	1.67	1.67	14.72	0.186	-8.52
	Window Newton	1.57	1.20	14.76	0.155	-8.50
Signed + risk + beta	Sequential	1.83	1.83	13.20	0.203	-8.44
	Window Newton	-2.69	-3.68	13.10	-0.221	-9.43

Table A.2: Sequential and window Newton portfolio metrics, corresponding to Table 2. Returns, volatility, and drawdown are percentages. NAV metrics use reconstructed daily states; CAGR, volatility, and Sharpe use the evaluated local net returns.

Comparison at a common depth. To distinguish windowing from the larger final depth allowance, we also compare the saved histories at depth at most 48: up to 48 daily batches or 16 three-day window batches. Each entry uses the last accepted path available within that allowance, with no selection based on its tracking error. Table A.3 reports the three long-short policies. At this common depth, windowing lowers NAV error in all three cases and return tracking error in one case. The equal-depth comparison therefore gives a more limited assessment than the final comparison at different depth allowances.

Policy	Daily NAV error	Window NAV error	Daily TE	Window TE
Long short	4.949	4.161	3.022	3.588
Signed + beta	4.783	4.391	3.319	3.081
Signed + risk + beta	4.563	2.537	2.752	3.037

Table A.3: Long-short tracking at a common policy-depth allowance of 48. All entries are percentages.